

ZINEDDINE REBBOUH

Machine Learning Engineer

[LinkedIn](#) | +213 562 163 159 | [Portfolio](#) | zinedinerabouh@gmail.com | [GitHub](#)
Touggourt, Algeria

Summary

- Machine Learning Engineer with an M.Sc. in Artificial Intelligence and Data Science and hands-on experience designing, training, and evaluating ML models end to end in Python, from data preprocessing and feature engineering through hyperparameter tuning, benchmarking, and deployment.
- Experienced with MongoDB, PostgreSQL, and Redis for data storage, querying, and retrieval, and with scikit-learn, PyTorch, and TensorFlow/Keras for building and validating classification, regression, clustering, forecasting, and recommendation models.
- Delivered a production AI platform (Tawfir) combining a RAG support pipeline, a hybrid recommendation engine, and a computer vision classifier, with rigorous evaluation (ROC AUC, Recall@k) and documented experiments for reproducibility.
- Comfortable working independently in remote, asynchronous, cross-functional teams, with strong analytical thinking, documentation habits, and a fast learning curve on new frameworks and datasets; M.Sc. thesis graded 19/20.

Technical Skills

- Programming:** Python (Advanced), JavaScript/TypeScript, SQL, Java
- Machine Learning:** scikit-learn, PyTorch, TensorFlow/Keras, XGBoost, Pandas, NumPy, GridSearchCV, hyperparameter tuning, model benchmarking
- Data & Databases:** MongoDB, PostgreSQL/PostGIS, pgvector, Redis, storage, querying, retrieval, preprocessing, feature engineering
- Generative AI & NLP:** RAG pipeline design (retrieval, chunking, reranking), embedding models, HuggingFace Transformers
- Computer Vision:** CNN, 3D CNN, ResNet-50 transfer learning, YOLOv8, MediaPipe
- MLOps & Deployment:** Docker, GitHub Actions CI/CD, FastAPI, Django REST Framework, Render, model serving and inference pipelines, Celery, monitoring; familiar with Kubernetes and MLflow
- Languages:** Arabic (Native), English (C1 Advanced, EF SET Certified), French (B1 Intermediate)

Work Experience

Machine Learning Engineer Intern	Elevvo Pathways	Remote (Cairo, Egypt)	Jul 2025 – Aug 2025
- Designed, trained, and evaluated 6 end-to-end ML pipelines (regression, classification, clustering, forecasting, recommender systems, and computer vision) within a 2-month remote engagement, benchmarking each model against baseline metrics. - Built an XGBoost sales-forecasting model using lag features, rolling statistics, and 5-fold GridSearchCV hyperparameter tuning; validated results with time-series cross-validation. - Trained a CNN for traffic-sign recognition with data augmentation and implemented a collaborative-filtering recommender, evaluated with precision@k and recall@k benchmarks. - Produced clean, documented, reproducible Python code and technical experiment write-ups while collaborating asynchronously with a fully remote, international team across multiple time zones.			

Key Projects

Tawfir, Production AI Platform for Food Waste Redistribution	M.Sc. Thesis, 2026
- Engineered a RAG-based AI support pipeline (Gemini 2.5 Flash, text-embedding-004, pgvector/HNSW) reaching 89% Recall@4 across 65 knowledge chunks with 68% autonomous ticket resolution and zero hallucinations across 30 evaluated conversations; also trained an AI freshness classifier via phased ResNet-50 transfer learning, reaching ROC AUC 0.9520. - Built a hybrid recommendation engine fusing content-based, collaborative, geospatial, and time-aware signals, using PostgreSQL/PostGIS for data storage and geospatial querying, rebuilt hourly via Celery background workers. - Deployed and monitored the platform on Render with Docker-based service packaging and GitHub Actions CI/CD across 6 two-week Agile sprints, achieving sub-600ms cold and sub-80ms cached response times via Redis caching.	
Football Analytics System, Real-Time Computer Vision Pipeline	2025/2026
- Built a real-time YOLOv8 + ByteTrack pipeline for multi-object detection and tracking of players, ball, referees, and goalkeepers, sustaining over 95% tracking ID consistency at 50–60 FPS. - Engineered automatic team assignment via K-means jersey-color clustering with temporal smoothing, and applied optical-flow camera compensation with homography-based pitch calibration to compute player speed and distance within 0.5 km/h and 2–3% error.	
Arabic Sign Language Detection, Real-Time Recognition System	2025/2026
Built an end-to-end recognition system using a custom 3D CNN (14.3M parameters), reaching 90.71% validation accuracy on a 20-class video dataset, with a MediaPipe-based preprocessing pipeline (hand detection, ROI extraction, CLAHE enhancement) and real-time webcam inference.	

Education

M.Sc. Artificial Intelligence and Data Science	University of Constantine 2	Algeria	2024 – 2026
Thesis: Tawfir, a production AI platform deployed on Render. Grade: 19/20. Coursework: Statistical Learning, Deep Learning, NLP, Optimization for ML, Data Mining, Big Data Systems.			
B.Sc. Computer Science	University of Constantine 2	Algeria	2021 – 2024
Graduation project ranked 1st in cohort, score 17/20 (full-stack multivendor e-commerce platform), defended in English.			

Certifications & Honours

- Machine Learning Specialization (Andrew Ng, DeepLearning.AI/Coursera); Natural Language Processing (OpenHPI/Hasso Plattner Institute); Understanding Data Engineering (DataCamp)